Clay Goldsmith

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Education

Rice University, Wiess School of Natural Sciences Houston, TX
Bachelor of Science (BS) in Physics (Computational Specialization), Minor in Engineering Design Expected Dec 2027

Professional & Research Experience

Undergrad Researcher, Rice University Bonner Nuclear Lab *August 2026 - Present*

- FPGA-driven data filter in the Large Hadron Collider's Level 1 Trigger for displaced beam halo muon tagging.
- C prototyping and simulation, direct implementation with Verilog and HSL on an FPGA dev board

Full Stack Software Engineering Intern, Campus *May 2026 - August 2026*

- Designed & built an internal code review agent
  - full 1000+ line typical PR review in 5-10 minutes
  - <$0.06 per review
  - 100-200 parallel specialized agents deployed during some steps of the review to complete short, well-defined tasks
- Designed & built a second agentic system to synchronize code review rules with legal source policy in real time
  - fast retroactive sweeps against code historically linked to policy when policy changes; auditable
  - automatic synthesis of 100s of brief policy source rules into ~40 codebase-wide broad criteria that meet obligations
- Wrote Terraform config to deploy systems using Scalr; configured security & PII; ran code-review bot on itself!

Student Research Assistant, Baylor High Energy Physics Lab *May 2024 - August 2024*

- Worked directly under Dr. Jay Dittmann on Baylor's CMS detector replica, collaborating with researchers from Baylor, Fermilab, and CERN on proposed Large Hadron Collider improvements.
- Sketched out ideas for anomalous data detection using Topological Data Analysis and Classifier neural networks.
- Designed, modeled, and manufactured a test stand for the detector's upgraded scintillator tile array using Onshape and a laser cutter.
- Rebuilt the lab's CMC card testing database from a legacy site to Python 3 and MongoDB, designed to handle test logs for ~30,000 cards ahead of a future (HGCAL) upgrade.

Project Experience

Software Lead, Rice Robotics R&D *September 2025 - Present*

- Leading RL gait training for a quadruped feline robot with a novel laser-pointer control interface. Built actor/critic fundamentals in a custom PyBullet/Gym sim before moving to Genesis for full training runs.
- Achieved consistent locomotion at ~0.8 m/s in simulation with generalized direction and orientation commands. Converges in under 2 hours (~1M iterations) thanks to Genesis.
- Training pipeline designed for real-world robustness: handles encoder loss and adapts to new physical configurations without total failure.
- Project management; teaching a team of 5 first-year CS students fundamentals of python, robotics, deep learning, simulation

Electronics Lead, Houston Zoo "CACTI" Cheetah Ball *August 2024 - May 2025*

- Led electronics design for a robotic enrichment toy built for the cheetahs at the Houston Zoo. 0.8m diameter, ~4kg, >2hr battery life, >100ft range
- Managed client communication with zoo staff directly
- Balancing durability/safety constraints against engineering tradeoffs across CAD, Arduino programming, and systems integration using objective metrics and regular meetings to refine target specs.

Python, Tensorflow, PyTorch, ROS2, Onshape, KiCAD, Machine Learning, Reinforcement Learning, Pybullet, Genesis, Sim-to-Real Transfer, Electrical Design, Docker, Raspberry Pi, Arduino, Embedded Systems, MongoDB, Linux/Ubuntu, Microsoft Office, Git, Cloudflare, Networking

Interests: Factorio, Running, Bouldering, Jazz Piano, Rock Guitar, Electronic Music, Martel College